

MACHINE LEARNING CANVAS

Iteration:

Designed for:

Designed by:

Date:

PREDICTION TASK  What is the type of task? Which entity are predictions made on? What are the possible outcomes to predict? When are outcomes observed?	DECISIONS  How are predictions turned into actionable recommendations or decisions for the end-user? (Mention parameters of the process / application for this.)	VALUE PROPOSITION  Who is the end beneficiary, and what specific pain points are addressed? How will the ML solution integrate with their workflow, and through which user interfaces?	DATA COLLECTION  How is the initial set of entities and outcomes sourced (e.g., database extracts, API pulls, manual labeling)? What strategies are in place to update data continuously while controlling cost and maintaining freshness?	DATA SOURCES  Where can we get data on entities and observed outcomes? (Mention internal and external database tables or API methods.)
IMPACT SIMULATION  What are the cost/gain values for (in)correct decisions? Which data is used to simulate pre-deployment impact? What are the criteria for deployment? Are there fairness constraints?	MAKING PREDICTIONS  Are predictions made in batch or in real time? How frequently? How much time is available for this (including featurization and decisions)? Which computational resources are used?		BUILDING MODELS  How many models are needed in production? When should they be updated? How much time is available for this (including featurization and analysis)? Which computation resources are used?	FEATURES  What representations are used for entities at prediction time? What aggregations or transformations are applied to raw data sources?
MONITORING  Which metrics and KPIs are used to track the ML solution's impact once deployed, both for end-users and for the business? How often should they be reviewed?				



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Filling in all 10 boxes is the easy part.

A filled canvas isn't a plan. It's a set of claims: what's predictable, what data exists, who will act on the prediction, and what that's worth. Every one of them can be wrong on a canvas that looks finished.

7 checks before you build:

- **A number, in dollars.** What one avoided incident or one retained customer is worth. Without it you can't set a deployment threshold or defend a budget.
- **A named person who acts on the prediction.** At a stated threshold, with capacity for the volume it produces.
- **Every feature available at the moment of prediction.** Anything you only know after the outcome is leakage. It inflates your offline results and nothing else.
- **Labels that exist.** Count the past cases whose outcome you already know, and how many of those are the case you're predicting. A case only counts once the wait window has closed: the time between making the prediction and seeing what actually happened.
- **A holdout, if acting on the prediction changes the outcome.** Send a retention offer to everyone the model flags and you never learn who would have stayed anyway. The next model trains on results your own actions produced. Keep a small share out of it.
- **An evaluation counted in events, not rows.** A test set of 35,040 machine-hours can contain 3 or 4 failures, and a recall figure measured on 4 failures isn't stable. Aim for at least 30.
- **Legible to someone who wasn't in the room.** Most of the people who use it later weren't.

Whatever you can't answer is the work left before anyone writes code.

"The Canvas allowed us to structure and focus our delivery sprints. The ML system went live in June 2020 and still is."

— Christian Schiller, Senior IT & Enterprise Architect, Deutsche Telekom

Get help running it

Louis Dorard created the canvas in 2015 and runs it as a working session on live projects: frame the problem, then test that framing against the data you actually have. You leave with a dollar estimate of what the system is worth, and a spec your engineers/agents can build from.

Bring a project you're scoping now, and you'll leave the call knowing which parts of your canvas your data won't support.

[Book a 30-minute call →](#)

Working on this with someone else? Send them to ownml.co/machine-learning-canvas for their own copy.

Reference the Machine Learning Canvas

If you need to reference the Canvas formally (e.g. in documentation or publications), you may use:

```
@misc{mlc,
  title = "Machine Learning Canvas",
  author = "Louis Dorard (OWNML)",
  url = "https://www.ownml.co/",
  year = "2015"
}
```

"MLC played an essential role in a positive project outcome. It is now being used throughout the organization."

— Daitan, now part of Coforge, a global IT services firm